MAE 199 Final Presentation: Occupancy Optimized VAV Control

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Problem statement and goals

- **Problem:** Currently the Center Hall and many other campus buildings run HVAC systems on occupied mode constantly from (6:30am to 10:30pm) regardless of real occupancy.
    - This schedule does not implement standby mode and wastes energy by running the HVAC when the building is not occupied.
- **Goal:** Optimizing the VAV control for occupancy to save energy through reduced AHU load and heating/cooling demand.
    - Controls to reduce the time spent in occupied mode should not significantly impact thermal comfort or facilities management operations.

Fall 2024 Work: Summarized

- Literature Review + Presentation
- Preliminary Control Design Strategies outlined in design document
- Dashboarded VAV damper position, occupancy, zone temperature, and Center hall AHU energy use on Grafana.
- Consolidated control strategies into 4 levels.
- Calculated potential energy savings based on AHU load reduction.
- Presented at Lab Expo in January 2025.

Reducing Energy Consumption of HVAC Systems Through Occupancy Optimized VAV Control

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Introduction

- Heating, Ventilation, and Cooling (HVAC) systems are responsible for 37% of the energy consumption of a typical university or educational institution building.
- HVAC systems being operated on inefficient schedules (i.e. running when spaces are unoccupied) waste thermal energy from heating/cooling and electricity, largely from fan loads.
- Many buildings at UCSD operate on a generous HVAC schedule of 6:30AM-10:30PM and do not rely on occupancy data to inform operations.
- **Variable Air Volume (VAV)** boxes allow for programmable control for air flow and conditioning within large buildings.

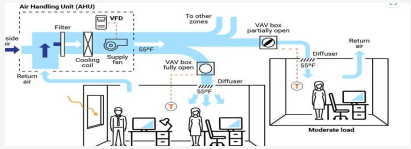

Figure 1: VAV Diagram

Proposed Solution

Implement a control method in 4 escalating levels which gradually tightens the HVAC schedule in order to better link operations to true occupied hours.

- By adjusting metrics such as the airflow and temperature band setpoints through set modes of occupancy, we hope to find savings for facilities management.
- **Higher levels of control** reflect a greater potential for savings with more difficult implementation.
- **Lower levels of control** maintain a steady baseline for that has the lowest potential for issues in implementation.

Objectives of Solution:

- Adherence to CA title 24 building code and ASHRAE guidelines.
- Maintain occupant comfort (CO_2, temperature, airflow).
- Increased energy savings from fan load and heating/cooling.

Control Levels: Each n level follows the n-1 level's rules.

Level 1	Mark room as Unoccupied mode for all hours of weekend, override if occupancy is detected.
Level 2	Tightened weekday Occupied mode hours. Ex: 6:30 to 8:30 vs 10:30. Full override to normal.
Level 3	Implemented time out clock to return to unoccupied after no motion is detected.
Level 4	Implement standby mode with direct link to occupancy sensors throughout weekdays. Ex: VAV 22

Methodology

- Building Automation Software (BAS) **METASYS** collects building information, such as light motion sensor activity and HVAC activity. This data is analyzed through visualization tool Grafana.
- We clearly define control modes for occupancy for the VAV boxes:

Occupied	Damper position of ~40% open, Temp Max 70°F, Temp Min 65°F
Unoccupied	Damper position of 0% open, Temp Max 90°F, Temp Min 55°F
Standby	Maintains intermediary temperature setpoints (not yet implemented)

- As we escalate, control methods will be tested and METASYS data will provide us with the opportunity to validate the effectiveness and any changes to air comfort.
- An important assumption is accurate occupancy data as the basis for our strategies.

Results & Discussion

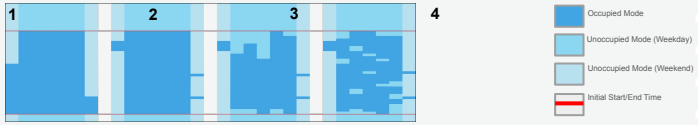

Figure 2: Visualizations of Levels 1, 2, 3, 4 of HVAC control.

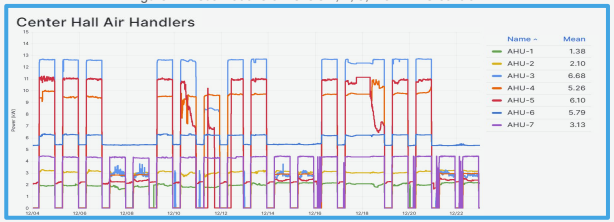

Figure 3: kW consumption of AHU 1-7 in Center Hall over the period 12/04/24-12/22/24.

- During evening hours from 10:30pm to 6:30am the AHU are set to unoccupied mode and VAVs are closed, on weekends only some of the AHU are set to unoccupied.
- **Level 1:** In an idealized projection, where all weekend hours are unoccupied, 453.85 kWh are saved. (Savings of $59/day for electric costs of $0.26/kWh).
- **Level 2:** If 2 hours each week day, 77.2 kWh are saved, ($20.07/day savings). For a total calendar year of weekends being unoccupied and a 2 hour reduction of occupied hours during the day, this results in a 16.38% decrease in electric consumption by the AHU.
- **Levels 3 and 4:** Implement intermediary modes and timeouts that are unpredictable, thus we consider ideal schedules to get a baseline for savings.
- Note that these **AHU calculations only consider the reduced AHU fan load from changing the VAV damper positions.** While harder to quantify, the thermal energy savings from the widened heating and cooling setpoints are just as significant if not more.

Case Study: Center Hall VAV 22

- Center Hall 3rd floor is ideal for testing due to pre-existing lighting sensors, air handler submetering data access / control via Metasys, and zone use type (office / classrooms).
- **VAV 22 is tied directly to room occupancy**, significantly reducing the amount of time it spends open (effectively level 3 control). Despite this VAV 22's temperature control was effectively equal to all other VAV operations.

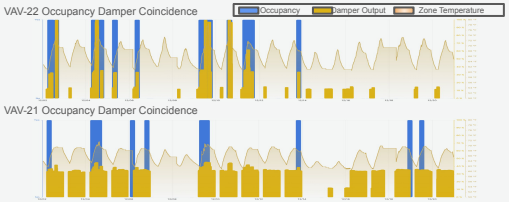

Figure 4: Room temperature, occupancy, and VAV damper position of VAV 21 and 22 from 12/02/24-12/20/24

Conclusion & Future Work

- VAV 21 & 22 case study fulfills all objectives of proposed solution, and demonstrates that tighter VAV control is possible.
- AHU fan load savings of roughly $59 per weekend day and $20 per weekday can be achieved through levels 1 and 2 of control.
- Additional savings come from reduced heating and cooling need.
- Shows the potential for good occupancy data to provide energy savings, supporting the investment into occupancy estimation.

Future Work:

- Pitch full plan to Facilities Management (FM) to gain permission to implement and test control schemes.
- Continuously monitor and assess METASYS data findings for impactful results and areas of improvement.
- Launch occupant comfort surveys in affected zones for additional validation and feedback.

References

Figure 1: https://www.pnnl.gov/projects/om-best-practices/variable-air-volume-systems
Figures 3 & 4: http://derconnect-brick.sdsc.edu:3000/d/fe12ed1b-1a5f-48d6-bd2f-4c8004c378e3/mae 199-center-hall-academic-senate?orgId=1&from=1732089600000&to=1736409599000

Acknowledgements

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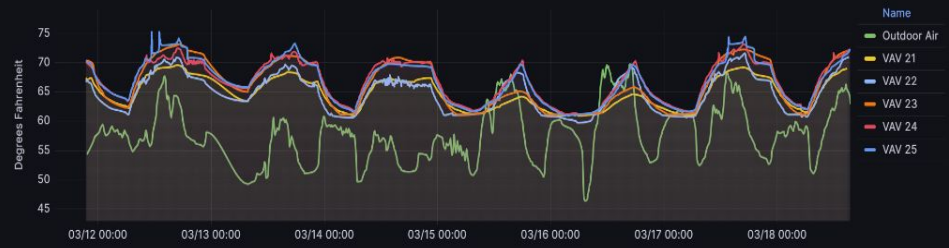
Zone Temperature



Zone Occupancy

VAV-21 | Supply Air Flow

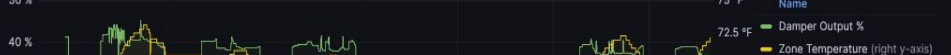
VAV-21 Damper Output

VAV-22 | Supply Air Flow

VAV-22 Damper Output

VAV-23 | Supply Air Flow

VAV-23 Damper Output



Control algorithm design

- Adapted from Shelly control PLC code developed for Brick Schema research.
- Utilizes current occupancy data along with implementing a forecast for better prediction.
- Runs over a schedule time and passes occupancy data through logic statements to determine whether to turn the plug load on or off.
- Links to Brick server to communicate with Metasys (BAS) to control building operations.
- Allows us to control VAV modes of occupied, unoccupied, and standby based on real occupancy and also run simulations based on historical occupancy data.

Control Strategies

Stage	With Forecast	Without Forecast
Start of Day	At 6:30am we precondition the space, occupied for 1 ½ hours after 6:30am	At 6:30am we precondition the space, occupied for 1 ½ hours after 6:30am
Daytime Operation	After 15 minutes of no occupancy system checks forecast, possible occupancy within 15 min>> standby, no possible occupancy>> unoccupied	After 15 minutes of no occupancy system goes to unoccupied mode
End of Day	Nighttime operation can be moved up from 10:30pm if there is no occupancy or forecast any amount of time before 10:30	Nighttime operation can be moved up from 10:30pm if there is no occupancy any amount of time before 10:30
Nighttime Operation	No change, system goes to unoccupied at 10:30pm until 6:30am the following day.	No change, system goes to unoccupied at 10:30pm until 6:30am the following day.

Code Modifications

- Updated output from PLC binary to flip VAV status to occupied, unoccupied, or standby.
- Implemented counter to track occupied time, unoccupied daytime hours, total unoccupied time, and standby time over the course of runtime.
- Created minute resolution, weekly occupancy forecast for Center Hall Third Floor from 83 days of occupancy data.
- Implements standby based on forecast prediction threshold during daytime hours. (threshold currently set to any possibility over 0)

Final product

- Code takes an occupancy dataset (or current data), and a forecast dataset as inputs
- We can set time intervals that can span days, weeks, or longer.
- Outputs per minute VAV status and a summary of time spent in each VAV mode over the entire simulation period.
- Enables comparison of times of year, different locations, and use of the forecast.
- Provides us with key insights in metrics such as daytime unoccupied hours and standby hours quantifying the difference between out control and the normal schedule, allowing us to see how much energy we can save.

Tests we can run

- We can select to run with the forecast and standby mode which switches the VAV to standby when there is a high enough probability of occupancy but no current occupancy detected.
- Without the forecast we read current occupancy data and switch to unoccupied after no occupancy has been detected for a certain period of time.
- We can also change parameters like the threshold time with no occupancy detected.

Simulation Results

1 month run

Start Date: 2025-02-05 10:00:00

End Date: 2025-03-05 10:00:00

Occupied time: 5960 minutes, 99.33 hours

Unoccupied time: 28298 minutes, 471.63 hours

Standby time: 6062 minutes, 101.03 hours

Daytime Unoccupied: 14618 minutes, 243.63 hours

Total time (mins): 40320, Total time (hours): 672.00

Simulation ran with forecast and standby

Total control hours: **344.66 hrs**

Occupied time with no control:

672-(471.63-243.63) = **444 hrs**

Overall, **14.8%** of total time in occupied &

15.03% in standby with control vs **66.1%** in occupied without.

Avg **8.701** more unoccupied hours per day

Simulation Results

Week: Wednesday Feb 26th - Wednesday March 5th

```
Start Date: 2025-02-26 06:00:00
End Date: 2025-03-05 21:00:00

Occupied time: 2370 minutes, 39.50 hours
Unoccupied time: 7446 minutes, 124.10 hours
Standby time: 1164 minutes, 19.40 hours
Daytime Unoccupied: 3996 minutes, 66.60 hours
Total time (mins): 10980, Total time (hours): 183.00
Simulation ran with forecast and standby

Start Date: 2025-02-26 06:00:00
End Date: 2025-03-05 21:00:00

Occupied time: 2370 minutes, 39.50 hours
Unoccupied time: 8610 minutes, 143.50 hours
Standby time: 0 minutes, 0.00 hours
Daytime Unoccupied: 5160 minutes, 86.00 hours
Total time (mins): 10980, Total time (hours): 183.00
Simulation ran without forecast and standby
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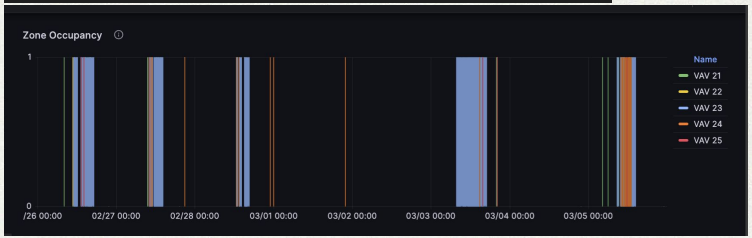

Day: Wednesday Feb 26th, 6am - 10:30pm

```
Start Date: 2025-02-26 06:00:00
End Date: 2025-02-26 22:30:00

Occupied time: 451 minutes, 7.52 hours
Unoccupied time: 417 minutes, 6.95 hours
Standby time: 122 minutes, 2.03 hours
Daytime Unoccupied: 387 minutes, 6.45 hours
Total time (mins): 990, Total time (hours): 16.50
Simulation ran with forecast and standby

Start Date: 2025-02-26 06:00:00
End Date: 2025-02-26 22:30:00

Occupied time: 451 minutes, 7.52 hours
Unoccupied time: 539 minutes, 8.98 hours
Standby time: 0 minutes, 0.00 hours
Daytime Unoccupied: 509 minutes, 8.48 hours
Total time (mins): 990, Total time (hours): 16.50
Simulation ran without forecast and standby
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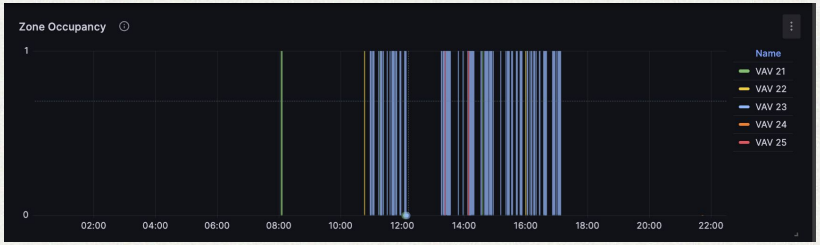
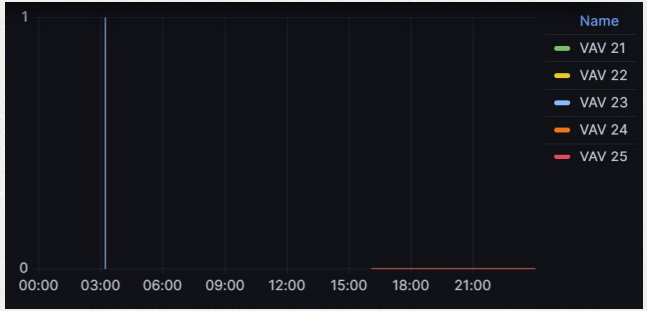


Simulation Results

Low Occupancy Day: February 8th 6:30am - 10:30pm

Occupied time: 106 minutes, 1.77 hours
Unoccupied time: 814 minutes, 13.57 hours
Standby time: 40 minutes, 0.67 hours
Daytime Unoccupied: 784 minutes, 13.07 hours
Total time (mins): 960, Total time (hours): 16.00
Simulation ran with forecast and standby

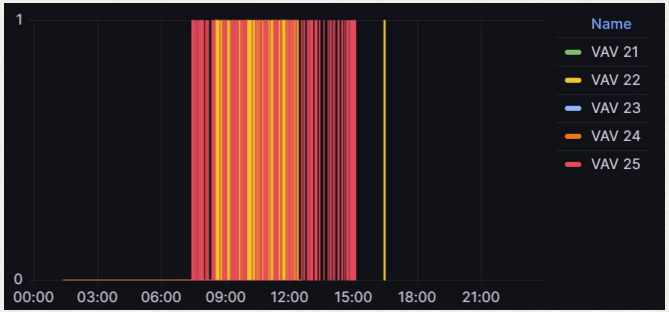
Date was a Saturday, forecast predicted almost no occupancy, resulting in most of the day being unoccupied



High Occupancy Day: March 12th 6:30am - 10:30pm

Occupied time: 548 minutes, 9.13 hours
Unoccupied time: 307 minutes, 5.12 hours
Standby time: 105 minutes, 1.75 hours
Daytime Unoccupied: 307 minutes, 5.12 hours
Total time (mins): 960, Total time (hours): 16.00
Simulation ran with forecast and standby

Wednesday with forecast. Heavy occupancy in the middle of the day but still significant reduction due to low evening activity and no forecast in later part of day.



Future work

- Present model to FM along with numbers from tests ran to show potential savings, implement true control test.
  - Collect ground truth occupancy to ensure data is accurately reflecting real occupancy.
  - Implement code into Brick Server and allow it to run off real time occupancy data.
- Implement temperature override/full override modes to control code.
- Update forecast model that auto loads historical occupancy data to improve prediction accuracy.
- Run control code separately for each VAV with its own forecast dataset

Questions?

Thank you!